Dynamic contrast-enhanced magnetic resonance imaging (MRI) and intravoxel incoherent motion imaging for characterizing angiogenesis in soft-tissue sarcomas: an exploratory study based on MRI-pathology co-alignment

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Background: Vascular endothelial growth factor (VEGF) expression and microvessel density (MVD) are established markers of tumor angiogenesis and key targets for anti-angiogenic therapy in soft-tissue sarcomas (STSs). Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) and intravoxel incoherent motion (IVIM) imaging are complementary magnetic resonance imaging (MRI) methods commonly used to assess angiogenesis in a variety of tumors. This study aimed to explore whether DCE-MRI and IVIM imaging could be used as non-invasive tools to evaluate tumor angiogenesis in STS based on MRI-pathology co-alignment.

Methods: In total, 70 patients with STS underwent 3.0-T DCE-MRI and IVIM imaging. The volume transfer constant (Ktrans), reflux rate constant (Kep), volume fraction of the extravascular extracellular space (Ve), true diffusion coefficient (D), pseudo-diffusion coefficient (D*), and perfusion fraction (f) were measured. Correlations between MRI parameters and VEGF expression or MVD were examined. Differences in MRI parameters between tumors with low and high VEGF expression were evaluated. Receiver operating characteristic (ROC) curves were generated, and interobserver reproducibility was assessed using the intraclass correlation coefficient (ICC). An MRI–pathology control protocol was implemented to verify the spatial correspondence between MRI slices and their corresponding histologic sections.

Results: Ktrans, Kep, D*, and f were positively correlated with VEGF expression (r=0.673, 0.573, 0.692, and 0.386, respectively) and MVD (r=0.662, 0.595, 0.619, and 0.344, respectively). Ktrans and D* were identified as independent factors associated with VEGF expression and MVD. No significant correlations were observed between Ve or D and either angiogenesis marker. Tumors with high VEGF expression exhibited higher Ktrans, Kep, D*, and f values than those with low VEGF expression. The areas under the ROC curves for the four parameters in differentiating between tumors with low and high VEGF expression were 0.861, 0.844, 0.860, and 0.720, respectively.

Conclusions: The study demonstrated associations between DCE-MRI- and IVIM-derived perfusion metrics and VEGF expression and MVD in STSs.

Keywords: Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI); intravoxel incoherent motion (IVIM); vascular endothelial growth factor (VEGF); microvessel density (MVD); soft-tissue sarcoma (STS)

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Introduction

Soft-tissue sarcomas (STSs) comprise a heterogeneous group of mesenchymal tumors and account for approximately 1% of all malignancies (1,2). STSs are often diagnosed at an advanced stage and are associated with high mortality rates. Angiogenesis is essential for tumor growth, invasion, and metastasis (3). Vascular endothelial growth factor (VEGF), a key regulatory cytokine, plays a pivotal role in regulating tumor angiogenesis (4). Microvessel density (MVD) is one of the most widely used methods for quantifying tumor angiogenesis (5). VEGF expression and MVD have been associated with clinical stage, recurrence, treatment response, and overall survival in patients with STS (6-10).

Accurate assessment of tumor angiogenesis is crucial for predicting tumor progression and monitoring the efficacy of anti-angiogenic therapy in STS, particularly in patients with metastatic disease (11-13). Currently, the preoperative pathological evaluation of STS requires needle biopsy, an invasive sampling procedure that is affected by multiple factors and may result in inappropriate treatment and suboptimal clinical outcomes (14). Non-invasive alternatives to histopathological indicators include quantitative parameters derived from magnetic resonance imaging (MRI) (15).

Multiple functional MRI technologies have been developed to evaluate intratumor heterogeneity in various cancers. Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) and intravoxel incoherent motion (IVIM) imaging are MRI techniques commonly used to characterize and quantify tissue perfusion. DCE-MRI uses an arterial input function and pharmacokinetic modeling to quantify the exchange of contrast agent between the intravascular and extravascular extracellular space (EES), providing information on tumor microcirculation, including tissue perfusion, capillary permeability, and vascular and interstitial volumes (16). IVIM imaging uses a biexponential model based on multiple b-values to distinguish between water molecule diffusion and microcirculation perfusion, without the administration of contrast medium (17). DCE-MRI and IVIM imaging have been applied in the assessment of tumor angiogenesis in glioma (18), ovarian cancer (19), cervical cancer (20), hepatocellular carcinoma (21), and pancreatic cancer (22).

The substantial intratumoral heterogeneity observed in STS underscores the need to precisely correlate the region of interest (ROI) on an image with the pathological sampling area. Previous studies have introduced MRI-pathology control methods to ensure accurate correspondence between MRI and pathology slices (23-26). Using this approach, several studies (25,26) have demonstrated the potential of IVIM imaging parameters to characterize microvascular features in a murine model of embryonal rhabdomyosarcoma. The present study aimed to explore whether DCE-MRI and IVIM imaging could be used as non-invasive tools to evaluate tumor angiogenesis in STS based on MRI-pathology co-alignment. We present this article in accordance with the STARD reporting checklist (available at https://qims.amegroups.com/article/view/10.21037/qims-2026-1-0086/rc).

Methods

Patients

The design of this prospective study was approved by the Ethics Committee of The Second Hospital of Dalian Medical University (No. KY2025-069-01). Informed consent was obtained from all patients. The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments.

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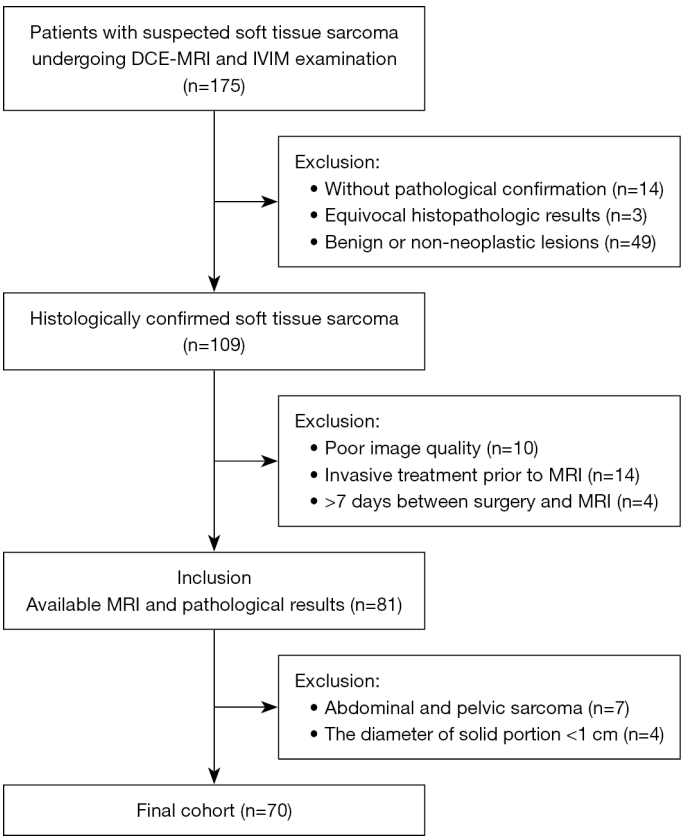


Figure 1 Study flowchart. DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; IVIM, intravoxel incoherent motion.

From July 2019 to October 2024, a total of 175 patients with suspected STS underwent multiple MRI examinations, including DCE-MRI and IVIM imaging. All patients were recruited using the convenience sampling method. Of these, 105 were excluded for the following reasons: (I) absence of pathological confirmation or equivocal histopathologic findings; (II) benign or non-neoplastic lesions; (III) poor image quality; (IV) invasive treatment before MRI; (V) an interval >7 days between MRI and surgery; (VI) abdominal or pelvic sarcomas; or (VII) a diameter of the solid portion of the sarcoma <1 cm. The study flow chart is shown in *Figure 1*.

MRI examination

All MRI examinations were performed on a 3.0-T scanner (Discovery MR750w, GE Healthcare) using the most suitable coil for each case (body, knee, or surface coil). Routine MRI protocols included axial T1-weighted fast spin-echo (FSE) sequences with a repetition time (TR)/echo time (TE) of 478–700/11–18 ms and an echo-train length of 3, as well as fat-suppressed T2-weighted FSE sequences with a TR/TE of 2,900–3,800/70–90 ms and an echo-train length of 19. Coronal and sagittal T1- and T2-weighted images were also acquired based on lesion location and morphology. The acquisition parameters for these sequences were as follows: slice thickness, 3 mm, slice gap, 1 mm, matrix, 384×384, field of view (FOV), 250–500 mm, and number of excitations (NEX), 2.

DCE-MRI data were acquired using T1-weighted fast spoiled gradient-recalled sequences with the following parameters: TR/TE, 5.0/2.0 ms; flip angle, 3°, 6°, 9°, and 12°; and slice thickness, 3 mm.

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The contrast agent was administered intravenously during the fourth dynamic acquisition using a power injector system (gadopentetate dimeglumine, 0.1 mmol/kg at 2 mL/s), followed by a 20-mL saline flush at 2 mL/s. The total scan time was 336 s (7 s for each phase, 48 phases). Delayed-enhanced images were acquired in three planes using T1-weighted FSE sequences.

Diffusion-weighted imaging (DWI) was acquired using spin-echo echo-planar imaging (SE-EPI) before the acquisition of contrast-enhanced T1-weighted images. The FOV, slice thickness, and slice gap for SE-EPI were identical to those of the conventional sequences. IVIM imaging was performed with a TR/TE of 3,000/70 ms; b-values of 0, 25, 50, 75, 100, 200, 500, and 800 s/mm^2; a matrix of 128×128; a NEX of 2–4; and a total scan time of 153 s. This set of b-values, including multiple low b-values, was selected to reliably separate perfusion and diffusion components, consistent with previous IVIM studies of STS (24,25).

MRI analysis

Quantitative analysis

Raw DCE-MRI and IVIM imaging data were imported into an Advantage Workstation (ADW 4.7, GE Healthcare) for processing. DCE-MRI data were analyzed using Gen IQ software, and Ktrans, Kep, and Ve maps were generated using the Tofts two-compartment model (16). Ktrans is the volume transfer constant that reflects the transfer rate of the contrast agent from the plasma to the EES. Kep is the reflux rate constant that reflects the rate at which the contrast agent returns to the plasma. Ve is the volume fraction of the EES that reflects the distribution volume of the contrast agent. For IVIM analysis, multi-b-value DWI data were processed using Functool-MADC software, followed by background noise reduction through intensity threshold adjustment. D, D*, and f maps were obtained from a biexponential model (17). D is the true diffusion coefficient that reflects the diffusion of water molecules outside the microcirculation. D* is the pseudo-diffusion coefficient that reflects the diffusion of water molecules in the microcirculation. f is the perfusion fraction that reflects the volume ratio of the perfusion effect to the total diffusion effect.

ROIs were independently placed by two radiologists (K.Z. and S.W. with 8 and 24 years of experience, respectively, in the radiological diagnosis of soft-tissue tumors) who were blinded to patients' clinicopathological data. Conventional MRI, DCE-MRI, and IVIM images were referenced to identify areas of significant enhancement and/or diffusion restriction in the tumor, while avoiding areas of hemorrhage, necrosis, and cystic changes. ROIs measuring approximately 50–80 mm^2 were placed in regions of homogeneous hyperintensity on the DWI map and then copied to the functional maps using 3Dsynchroview (GE Healthcare). The mean values of the measurements obtained by the two radiologists were used for the statistical analysis.

Qualitative analysis

Two radiologists independently assessed the MR images and evaluated the following parameters: (I) tumor size, defined as the longest measurable diameter of the tumor; (II) tissue layer, defined as the depth of the tumor relative to the superficial fascia, categorized as deep or superficial; (III) tumor margin, defined as the relationship between the tumor and adjacent surrounding structures, categorized as poorly defined or well-defined; (IV) tumor configuration, defined as the macroscopic contour and shape of the tumor, categorized as multilobulated or non-multilobulated; (V) tumor necrosis, defined as an area that was hyperintense on T2-weighted imaging (T2WI) without enhancement on contrast-enhanced images, categorized as <50% or ≥50%; (VI) T2WI tumor heterogeneity, defined as the presence of more than two types of signal intensities on T2WI, categorized as <50% or ≥50%; (VII) peritumoral edema, defined as fluid-like high signal intensity surrounding the tumor on T2WI that was clearly distinguishable from the

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solid components of the tumor, categorized as present or absent; and (VIII) peritumoral enhancement, defined as contrast uptake on post-gadolinium T1-weighted images, extending beyond the visible tumor margin, categorized as either present or absent.

MRI-pathology control method

An MRI-pathology control protocol was employed to confirm spatial alignment between MRI images and their corresponding histologic sections. This method, which has been used in animal and clinical studies of STS (23-27), comprises five sequential steps (*Figure 2*): (I) determining the location of the mass by palpation or ultrasound, drawing a positioning line (*Figure 2A*) on the surface perpendicular to the long axis of the body at the most palpable part of the mass, and arranging three vitamin AD capsules (P1, P2, and P3) equidistantly along line A; (II) performing a sagittal or coronal T2WI (*Figure 2B*) to assess the extent of the mass and the positions of the capsules, adjusting the axial imaging plane to obtain axial images of all three capsules (*Figure 2C,2D*), and then performing post-processing of the functional sequence to generate parameter maps (*Figure 2E*); (III) exposing the mass during surgery, placing three black non-absorbable sutures B, C, and D along positioning line A on the surface and placing a suture E above the mass (*Figure 2F*) to assist pathologists in determining the location of the mass during pathological sampling; (IV) removing the mass and fixing it in formalin, creating a pathological section corresponding to the MRI image (*Figure 2G-2I*); and (V) printing the DWI image containing the ROI at a scale corresponding to the actual size of the tumor (*Figure 2J*), completely removing the ROI from the printed image (*Figure 2K*), matching it with the pathological section (*Figure 2L,2M*), obtaining pathological sampling slices corresponding to the ROI (*Figure 2N*), and performing hematoxylin and eosin, VEGF, and CD34 staining (*Figure 2O-2Q*).

Pathological data acquisition and analysis

In accordance with the MRI-pathology control method, a paraffin block for immunohistochemical staining of VEGF and CD34 (a marker of MVD) was prepared from the solid portion of the formalin-fixed surgical specimen corresponding to the ROI on the MR image. Two pathologists (Y.L. and J.T. with 4 and 16 years of experience in musculoskeletal pathology diagnosis, respectively), who were blinded to the patients' clinicopathological data, provided histopathologic diagnoses of all specimens according to the World Health Organization classification of STSs [2020) (28).

VEGF expression was quantified and scored according to the method described by Han *et al.* (27). The percentage of positive cells at high magnification (×400) was scored as: 0 points, negative; 1 point, ≤10%; 2 points, 11–50%; 3 points, 51–75%; and 4 points, >75%. Staining intensity was classified as: 0 points, no staining; 1 point, light yellow; 2 points, brownish yellow; and 3 points, brown. The percentage score for positive cells and staining-intensity score were multiplied to obtain the immunoreactive score. The VEGF staining score was calculated as the mean score across five fields of view. VEGF expression was categorized as low (<4 points) or high (≥4 points).

MVD was measured by counting blood vessels using the method described by Weidner *et al.* (29). CD34-stained sections were initially examined at low magnification (×40) to locate three vascular "hotspots" exhibiting the highest vessel density. MVD was then quantified at high magnification (×200) by counting vessels in each hotspot, and the mean of the three counts was recorded. A single countable vessel was defined as any positively stained endothelial cell or cluster of endothelial cells. Blood vessels with a thick muscle layer or a diameter greater than eight red blood cells were excluded.

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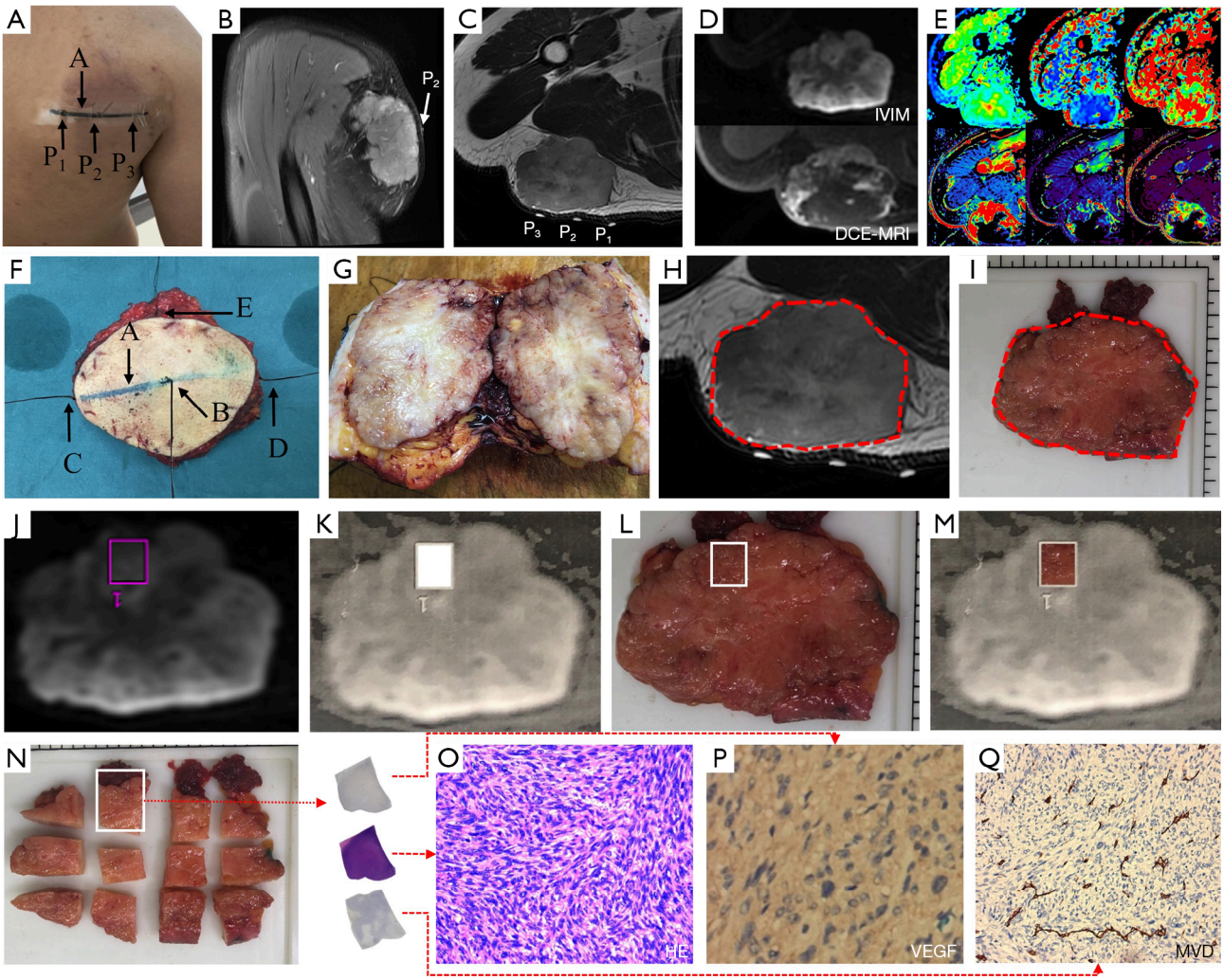


Figure 2 MRI-pathology control method. (A) Tumor localization using three vitamin AD capsules (P1, P2, and P3). (B) Sagittal T2-weighted image. (C) Axial T2-weighted image containing the three capsules. (D) Raw IVIM and DCE-MRI images. (E) Post-processed parametric maps of IVIM and DCE-MRI. (F) Surgical tumor specimen marked with orientation sutures. (G) Gross photograph of the resected tumor. (H,I) Red dashed outlines marked the tumor boundary for radiological-pathological matching. (J) DWI image containing the ROI. (K) Printed DWI image with the ROI excised. (L-N) The gross tumor specimen was matched to the coregistered DWI; the white rectangle denoted the tissue block corresponding to the MRI ROI for targeted pathological sampling. (O) HE staining (magnification, ×200). (P) IHC staining for VEGF (magnification, ×400). (Q) IHC staining for CD34 (magnification, ×200) for MVD assessment. P1, P2, and P3: three vitamin AD capsules serving as external fiducial markers for MRI localization. DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; DWI, diffusion-weighted imaging; HE, hematoxylin-eosin; IHC, immunohistochemistry; IVIM, intravoxel incoherent motion; MRI, magnetic resonance imaging; MVD, microvessel density; ROI, region of interest; VEGF, vascular endothelial growth factor.

Statistical analysis

The statistical analyses were performed using MedCalc version 15.8 (MedCalc, Mariakerke, Belgium) and SPSS version 25.0 (SPSS, Chicago, IL, USA). The normality of the data distribution was assessed using the Kolmogorov-Smirnov test, and homogeneity of variances was evaluated using Levene's test. Interobserver reliability was assessed using the intraclass correlation coefficient (ICC), with ICC values

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>0.75 considered indicative of strong agreement. Spearman correlation analysis was used to examine the correlations between the MRI parameters and markers of angiogenesis. The correlation coefficient (r) was defined as follows: <0.3, weak or no correlation; 0.3–0.7, moderate correlation; and >0.7, strong correlation. Multiple linear regression analysis was used to identify MRI parameters that were independently associated with markers of angiogenesis. An independent samples t-test or Wilcoxon rank-sum test was used to compare differences between tumors with high and low VEGF expression. Categorical variables were compared using the Chi-squared test. Receiver operating characteristic (ROC) curves were generated to evaluate the diagnostic performance of the MRI parameters in differentiating between tumors with high and low VEGF expression. The Youden index was calculated to determine optimal cutoff values, along with the area under the curve (AUC), sensitivity, specificity, and accuracy. Binary logistic regression analysis was used to construct a combined model and evaluate its diagnostic efficacy. AUCs were compared using the DeLong test. A P value <0.05 was considered statistically significant.

Results

Demographic and quantitative pathological characteristics

The study included 70 patients with STS, of whom 48.6% were male and 51.4% were female. The mean age of the patients was 59 years. The most commonly involved anatomical location was the thigh (n=31), followed by the trunk (n=15), upper limb (n=9), distal lower limb (n=5), shoulder (n=5), hip (n=4), and hand (n=1). Histopathologic diagnoses for all 70 patients are shown in *Table 1*. Representative MR images and VEGF and MVD maps of histologically confirmed myxoid liposarcoma and dermatofibrosarcoma protuberans are shown in *Figures 3*,*4*. The mean VEGF score was 4.73±2.15 (range, 0.50–10.30). Forty-four patients had tumors with high VEGF expression, while 26 had tumors with low VEGF expression. The mean MVD was 73.99±45.41 (range, 12.67–202.17). Table S1 summarizes VEGF expression and MVD in STS subtypes with more than five enrolled cases. The ICC values for VEGF, MVD, and MRI parameters were >0.75, indicating excellent interobserver agreement (*Table 2*).

Table 1 Histopathologic diagnoses of tumors with low and high VEGF expression

Histopathologic diagnosis	Values, n
Low VEGF expression	26
Myxoid liposarcoma	1
Dedifferentiated liposarcoma	2
Leiomyosarcoma	3
Angiosarcoma	1
Pleomorphic liposarcoma	1
Rhabdomyosarcoma	1
Fibrosarcoma	3
Well-differentiated liposarcoma	3
Malignant peripheral nerve sheath tumor	2

Table 1 *(continued)*

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Table 1 *(continued)*

Histopathologic diagnosis	Values, n
Undifferentiated pleomorphic sarcoma	1
Dermatofibrosarcoma protuberans	4
Synovial sarcoma	4
High VEGF expression	44
Myxoid liposarcoma	6
Synovial sarcoma	4
Leiomyosarcoma	10
Angiosarcoma	2
Fibrosarcoma	6
Rhabdomyosarcoma	1
Dedifferentiated liposarcoma	2
Well-differentiated liposarcoma	1
Malignant peripheral nerve sheath tumor	3
Undifferentiated pleomorphic sarcoma	5
Dermatofibrosarcoma protuberans	1
Alveolar soft part sarcoma	1
Extraskeletal myxoid chondrosarcoma	1
Malignant giant cell tumor of tendon sheath	1

VEGF, vascular endothelial growth factor.

Correlation analysis

Ktrans, Kep, D*, and f were moderately positively correlated with VEGF expression (correlation coefficients, 0.673, 0.573, 0.692, and 0.386, respectively; P=0.001 for f and P<0.001 for the other parameters) and MVD (correlation coefficients, 0.662, 0.595, 0.619, and 0.344, respectively; P=0.004 for f and P<0.001 for the other parameters). No significant correlations were observed between Ve or D and VEGF expression or MVD (all P>0.05) (*Table 3* and *Figure 5*). Multiple linear regression analysis indicated that Ktrans and D* were independently associated with VEGF expression (P=0.013 and 0.002, respectively) and MVD (P=0.004 and 0.045, respectively) (*Table 4*). D* and f were moderately positively correlated with Ktrans and Kep (r=0.621 and 0.569 for D*, respectively; both P<0.001; r=0.385 and 0.369 for f, respectively; both P<0.05) (*Table 5*). Subgroup analyses according to histologic subtype were not feasible because of the small sample sizes for individual STS subtypes.

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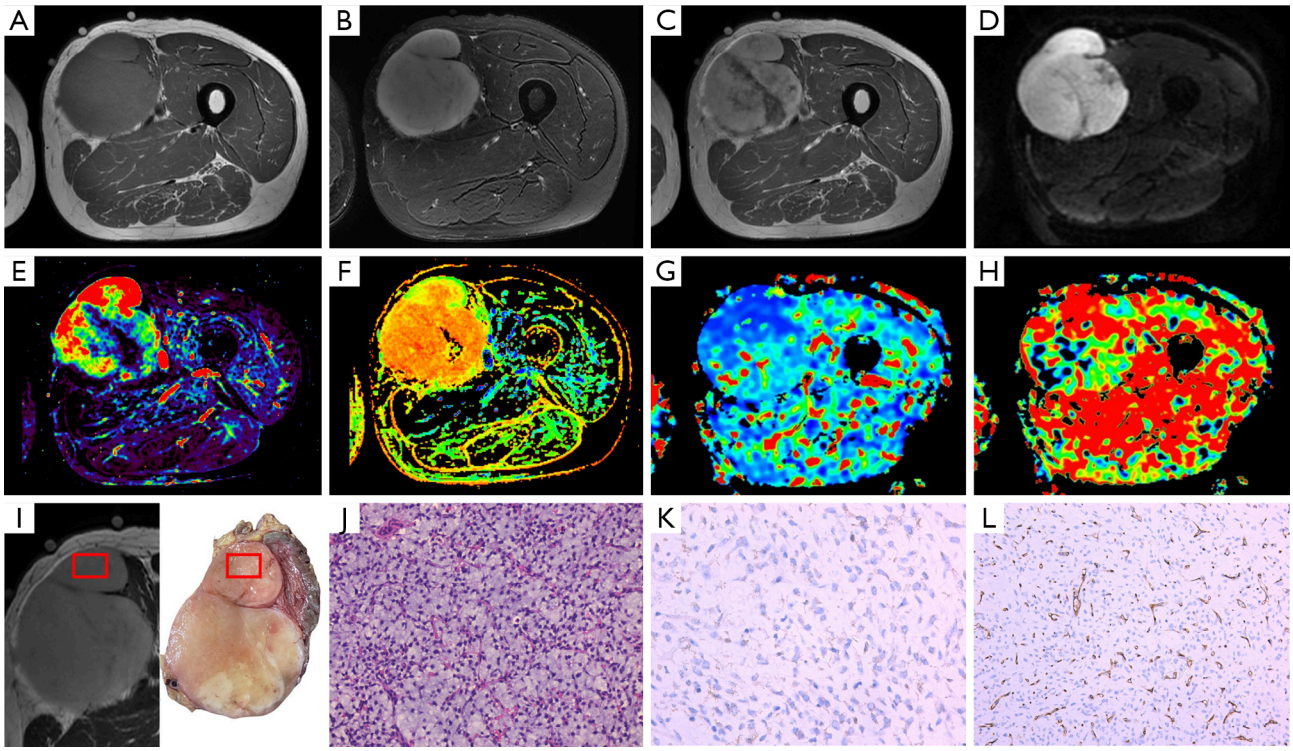


Figure 3 A 67-year-old patient with myxoid liposarcoma of the left thigh. The tumor showed slight hypointensity on the T1-weighted image (A) and homogenous hyperintensity on the T2-weighted image (B). The tumor showed significant heterogeneous enhancement after injection of a gadolinium-based contrast agent (C). The tumor showed slight heterogeneous hyperintensity on the IVIM (b =100 s/mm2) image (D). Ktrans (E), Kep (F), D* (G), and f (H) were 1.29 min−1, 1.32 min−1, 40.4×10−3 mm2/s, and 0.42, respectively. The pathological section corresponding to the MRI image showed the tumor was gray-white with a fish-fleshed appearance (I). HE staining (magnification, ×200) demonstrated tumor cells with round to oval morphology, accompanied by a frequently myxoid stroma, characteristic plexiform and branching vascular patterns, and variable numbers of lipoblasts (J). IHC staining for VEGF (magnification, ×400) showed a VEGF score of 8 (K). IHC staining for CD34 (magnification, ×200) showed an MVD of 124 (L). Red boxes indicate the imaging region of interest and its matched pathological sampling area. D*, pseudo-diffusion coefficient; f, perfusion fraction; HE, hematoxylin-eosin; IHC, immunohistochemistry; IVIM, intravoxel incoherent motion; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; MVD, microvessel density; VEGF, vascular endothelial growth factor.

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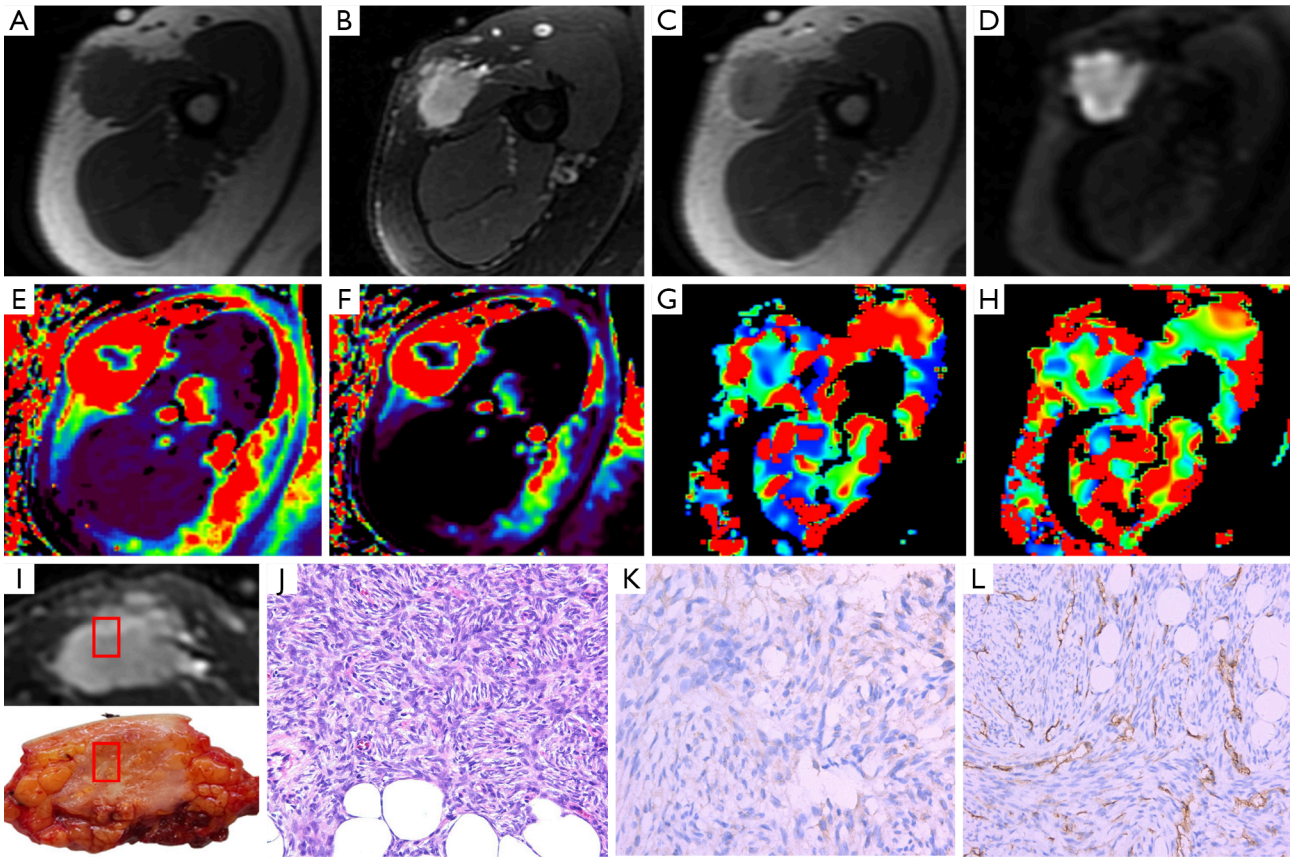


Figure 4 A 35-year-old patient with dermatofibrosarcoma protuberans of the right upper limb. The tumor showed hypointensity on the T1-weighted image (A) and homogeneous hyperintensity on the T2-weighted image (B), with infiltration of the surrounding fat plane. The tumor showed slight heterogeneous enhancement after injection of gadolinium-based contrast agent (C). The tumor showed slight heterogeneous hyperintensity on the IVIM (b =500 s/mm2) image (D). Ktrans (E), Kep (F), D* (G), and f (H) were 1.05 min−1, 1.26 min−1, 22.2×10−3 mm2/s, and 0.28, respectively. The pathological section corresponding to the MRI image showed the tumor was gray-white with a fish-fleshed appearance (I). HE staining (magnification, ×200) showed monotonous short spindle-shaped cells infiltrating adipose tissue in fascicular and herringbone arrangements, with mild focal cytologic atypia and increased mitotic activity (J). IHC staining for VEGF (magnification, ×400) showed a VEGF score of 4 (K). IHC staining for CD34 (magnification, ×200) showed an MVD of 68 (L). Red boxes indicate the imaging region of interest and its matched pathological sampling area. D*, pseudo-diffusion coefficient; f, perfusion fraction; HE, hematoxylin-eosin; IHC, immunohistochemistry; IVIM, intravoxel incoherent motion; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; MVD, microvessel density; VEGF, vascular endothelial growth factor.

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Table 2 Inter-reader ICCs for MRI parameters and angiogenesis markers

Parameters	Reader 1	Reader 2	ICC	95% CI
VEGF	4.74±2.22	4.72±2.18	0.914	0.865–0.946
MVD	78.31±48.49	69.69±44.12	0.915	0.867–0.946
Ktrans (min^{-1})	1.00±0.51	0.99±0.50	0.986	0.977–0.991
Kep (min^{-1})	1.29±0.67	1.31±0.68	0.918	0.871–0.948
Ve	0.74±0.21	0.73±0.20	0.864	0.790–0.913
D ($\times10^{-3}$ mm^2/s)	1.25±0.41	1.22±0.40	0.956	0.928–0.972
D* ($\times10^{-3}$ mm^2/s)	27.48±14.53	25.88±13.34	0.894	0.835–0.933
f	0.21±0.10	0.21±0.11	0.841	0.756–0.898

CI, confidence interval; D, true diffusion coefficient; D*, pseudo-diffusion coefficient; f, perfusion fraction; ICC, intraclass correlation coefficient; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; MVD, microvessel density; VEGF, vascular endothelial growth factor; Ve, volume fraction of the extravascular extracellular space.

Table 3 Correlations between MRI parameters and angiogenesis markers

Parameters	VEGF score		MVD	
	r	P	r	P
Ktrans (min^{-1})	0.673	<0.001*	0.662	<0.001*
Kep (min^{-1})	0.573	<0.001*	0.595	<0.001*
Ve	0.210	0.080	0.169	0.163
D ($\times10^{-3}$ mm^2/s)	−0.062	0.607	−0.004	0.972
D* ($\times10^{-3}$ mm^2/s)	0.692	<0.001*	0.619	<0.001*
f	0.386	0.001*	0.344	0.004*

, P value was statistically significant. D, true diffusion coefficient; D, pseudo-diffusion coefficient; f, perfusion fraction; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; MVD, microvessel density; VEGF, vascular endothelial growth factor; Ve, volume fraction of the extravascular extracellular space.

Comparative analysis

Tumors with high VEGF expression were more frequently associated with peritumoral edema and a multilobulated configuration than those with low VEGF expression (P=0.034 and 0.030, respectively). No significant differences were observed in the other qualitative parameters between tumors with high and low VEGF expression. Ktrans, Kep, D*, and f values were significantly higher in tumors with high VEGF expression than in those with low VEGF expression (P=0.002 for f and P<0.001 for the other parameters), while no significant differences were observed in Ve or D (P=0.246 and 0.738, respectively) (*Table 6*).

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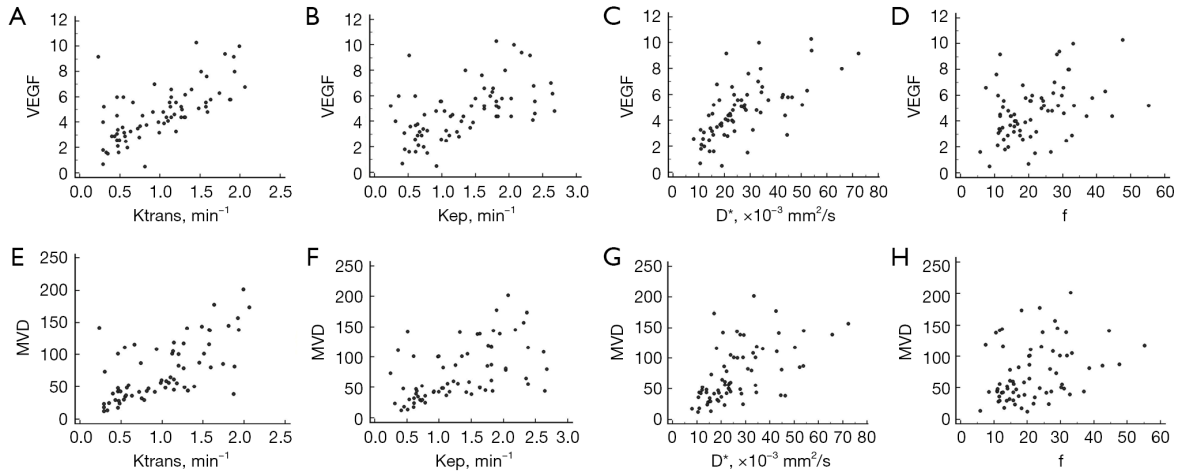


Figure 5 Scatter plots showing the correlations between Ktrans, Kep, D*, or f with VEGF expression (A-D) or MVD (E-H). D*, pseudo-diffusion coefficient; f, perfusion fraction; Kep, reflux rate constant; Ktrans, volume transfer constant; MVD, microvessel density; VEGF, vascular endothelial growth factor.

Table 4 Multiple linear regression analysis of MRI parameters independently associated with angiogenesis markers

Parameters	Angiogenesis markers	β coefficient	Standardized β coefficient	95% CI	P value
Ktrans (min^{-1})	VEGF	1.584	0.376	0.345 to 2.824	0.013*
	MVD	41.501	0.467	13.487 to 69.515	0.004*
Kep (min^{-1})	VEGF	0.172	0.053	–0.701 to 1.046	0.695
	MVD	2.786	0.041	–16.963 to 22.535	0.779
D* ($\times10^{-3}$ mm^2/s)	VEGF	0.063	0.399	0.305 to 2.269	0.002*
	MVD	0.916	0.274	0.020 to 1.812	0.045*
f	VEGF	–0.002	–0.012	–0.048 to 0.043	0.913
	MVD	–0.184	–0.041	–1.207 to 0.839	0.721

, P value was statistically significant. CI, confidence interval; D, pseudo-diffusion coefficient; f, perfusion fraction; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; MVD, microvessel density; VEGF, vascular endothelial growth factor.

Table 5 Correlations between DCE-MRI and IVIM imaging parameters

Parameters	Ktrans (min^{-1})		Kep (min^{-1})		Ve	
	r	*P*	r	P	r	P
D ($\times10^{-3}$ mm^2/s)	–0.089	0.466	–0.248	0.038*	0.177	0.142
D* ($\times10^{-3}$ mm^2/s)	0.621	<0.001*	0.569	<0.001*	0.179	0.139
f	0.385	0.001*	0.369	0.002*	0.151	0.213

, P value was statistically significant. D, true diffusion coefficient; D, pseudo-diffusion coefficient; DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; f, perfusion fraction; IVIM, intravoxel incoherent motion; Kep, reflux rate constant; Ktrans, volume transfer constant; Ve, volume fraction of the extravascular extracellular space.

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Table 6 MRI parameters in tumors with low and high VEGF expression

Parameters	Low VEGF expression (n=26)	High VEGF expression (n=44)	P value
Tumor size	6.79±3.50	7.60±5.63	0.512
Tissue layer			0.580
Deep	13 [50]	25 [57]	
Superficial	13 [50]	19 [43]	
Tumor margin			0.634
Poorly defined	8 [31]	16 [36]	
Well defined	18 [69]	28 [64]	
Tumor configuration			0.030*
Multilobulated	9 [35]	27 [61]	
Non-multilobulated	17 [65]	17 [39]	
Tumor necrosis			0.388
≥50%	14 [54]	19 [43]	
<50%	12 [46]	25 [57]	
T2WI tumor heterogeneity			0.193
≥50%	10 [38]	24 [55]	
<50%	16 [62]	20 [45]	
Peritumoral edema			0.034*
Present	11 [42]	30 [68]	
Absent	15 [58]	14 [32]	
Peritumoral enhancement			0.621
Present	14 [54]	21 [48]	
Absent	12 [46]	23 [52]	
Ktrans (min^{-1})	0.59±0.23	1.23±0.48	<0.001*
Kep (min^{-1})	0.81±0.30	1.59±0.64	<0.001*
Ve	0.69±0.15	0.76±0.22	0.246
D ($\times10^{-3}$ mm^2/s)	1.26±0.41	1.22±0.40	0.738
D* ($\times10^{-3}$ mm^2/s)	17.70±7.87	31.99±13.50	<0.001*
f	0.17±0.06	0.24±0.09	0.002*

Data are presented as mean ± standard deviation or n [%]. *, P value was statistically significant. D, true diffusion coefficient; D*, pseudo-diffusion coefficient; f, perfusion fraction; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; T2WI, T2-weighted imaging; Ve, volume fraction of the extravascular extracellular space; VEGF, vascular endothelial growth factor.

ROC curve analysis

The diagnostic efficiency of the perfusion parameters and combined models in differentiating between tumors with low and high VEGF expression is summarized in *Table* 7 and *Figure 6*. In the ROC curve analysis, Ktrans showed the highest AUC (0.861). The AUC of D* was significantly higher than that of f (0.860 *vs.* 0.720, P=0.024). No significant differences were observed in the AUCs of the other parameters (P>0.05). The combined model based on DCE-MRI-derived Ktrans and Kep achieved an AUC of 0.868, with a sensitivity of 84.1% and a specificity of 88.5%. The combined IVIM imaging parameters (D* and f) yielded an AUC of 0.870, with a sensitivity of 90.9% and a specificity of 73.1%. The comprehensive model integrating DCE-MRI and IVIM imaging parameters achieved an AUC of 0.900, with a sensitivity

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of 86.4% and a specificity of 84.6%. No significant differences were observed among the AUCs of the three combined models.

Table 7 Diagnostic performance of MRI parameters and combined models for differentiating between tumors with low and high VEGF expression

Parameters	AUC	95% CI	Cutoff value	Youden index	Sensitivity (%)	Specificity (%)	Accuracy (%)
Ktrans (min^{-1})	0.861	0.758–0.932	0.83	0.703	81.8	88.5	84.3
Kep (min^{-1})	0.844	0.737–0.920	1.25	0.650	72.7	92.3	80.0
D* ($\times 10^{-3}$ mm^2/s)	0.860	0.756–0.931	19.95	0.656	88.6	76.9	84.3
f	0.720	0.600–0.821	0.22	0.430	54.6	88.5	67.1
DCE-MRI	0.868	0.765–0.937	–	0.726	84.1	88.5	85.7
IVIM	0.870	0.768–0.938	–	0.640	90.9	73.1	84.3
DCE-MRI + IVIM	0.900	0.805–0.959	–	0.710	86.4	84.6	85.7

AUC, area under the curve; CI, confidence interval; D*, pseudo-diffusion coefficient; DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; f, perfusion fraction; IVIM, intravoxel incoherent motion; Kep, reflux rate constant; Ktrans, volume transfer constant; MRI, magnetic resonance imaging; VEGF, vascular endothelial growth factor.

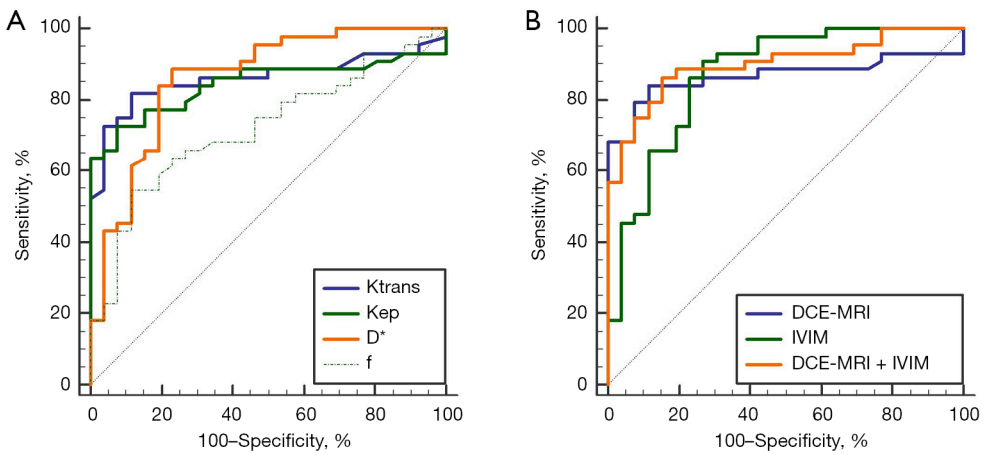


Figure 6 ROC curve showing the diagnostic performance of the quantitative parameters and combined models. (A) ROC curves for Ktrans, Kep, D* and f. (B) ROC curves for DCE-MRI, IVIM imaging, and DCE-MRI + IVIM imaging. D*, pseudo-diffusion coefficient; DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; f, perfusion fraction; IVIM, intravoxel incoherent motion; Kep, reflux rate constant; Ktrans, volume transfer constant; ROC, receiver operating characteristic.

Discussion

Spatially matched associations between imaging and histopathology metrics are crucial for radiologic-pathologic comparison studies in tumors. The present study explored the relationship between imaging parameters and tumor microvascular characteristics using an MRI-pathology co-registration method. The findings demonstrated that Ktrans and Kep derived from DCE-MRI, as well as D* and f derived from IVIM imaging, have considerable potential for evaluating tumor angiogenesis in STS.

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Tumor vasculature is predominantly composed of immature neovessels consisting of endothelial cells and basement membranes with incomplete structures, which can affect the hemodynamics of the intratumoral microcirculation. Neovascularization provides tumors with oxygen and nutrients, and eliminates metabolic waste. VEGF and MVD are commonly used markers for assessing tumor angiogenesis and key targets for anti-angiogenic therapy in STS (12,13). VEGF functions as a mitogen and pro-angiogenic factor by stimulating endothelial cell migration and proliferation, promoting neovascularization, and increasing vascular permeability. MVD is used to quantitatively evaluate the number of microvessels. Previous studies have reported that VEGF overexpression and MVD are significantly associated with poor prognosis in patients with STS (6-8).

Conventional MRI sequences remain essential for radiologists in the preoperative evaluation of STS. In the present study, tumors with high VEGF expression were more frequently associated with peritumoral edema and a multilobulated configuration. Several studies have demonstrated that these two imaging signs may be useful for predicting pathological outcomes of STS. Crombé *et al.* reported that peritumoral edema was associated with high-grade STS (30). Li *et al.* found that a multilobulated configuration was also associated with high-grade STS (31) and was useful for evaluating HIF-1α expression (32). Sedaghat *et al.* reported that a multilobulated configuration was predictive of tumor recurrence in STS (33,34).

Our findings suggest that the presence of these two signs may provide preliminary indications of high angiogenic activity in STS. It may be that VEGF promotes the formation of distorted, dilated, and haphazardly branched tumor microvessels with heterogeneous intratumoral distribution, leading to inconsistent growth rates in different regions and the loss of regular tumor configuration. In addition, VEGF significantly increases microvascular permeability, resulting in massive extravasation of plasma and proteins into the peritumoral interstitium. Combined with increased intratumoral interstitial pressure and impaired lymphatic drainage, this process may lead to prominent peritumoral edema.

DCE-MRI can quantify the transfer of contrast agents and has been widely used to assess tumor angiogenesis across various cancers (19-21). Ktrans and Kep have been shown to be useful for tumor diagnosis (35), grading (36), and hypoxia assessment (32) in STS. In the present study, Ktrans and Kep were moderately positively correlated with VEGF expression and MVD, consistent with previous findings in hepatocellular carcinoma (21), lung cancer (37), and chondrosarcoma (38). Ktrans and Kep also differed significantly between tumors with low and high VEGF expression. Ktrans and Kep are mainly influenced by capillary permeability, tissue blood flow, and the surface area. Increased VEGF stimulates extensive angiogenesis and increases blood flow, while immature neovascularization increases vascular permeability, facilitating a more rapid and increased exchange of contrast agents between the two compartments, ultimately leading to elevated Ktrans and Kep values.

No significant correlations were observed between Ve and VEGF expression or MVD. Ve is affected by tumor tissue necrosis, cellular density, and the extracellular matrix, which may limit its ability to accurately reflect alterations in the EES associated with tumor angiogenesis. A previous study (39) demonstrated that Ve exhibited considerable instability due to the influence of surrounding tissue edema. Studies of different tumor types have yielded inconsistent results regarding the relationship between DCE-MRI parameters and markers of angiogenesis. One study (19) found inverse correlations between Ktrans or Ve and VEGF expression in epithelial ovarian cancer, with similar findings reported in head and neck cancer (40). The inconsistent findings between the present study and previous studies may be explained by differences in the degree of vascularization or the maturity of neovasculature among different types of tumors, mismatches between the ROI and histopathological sampling locations, or differences in pharmacokinetic models and contrast agents.

IVIM can simultaneously provide information on tissue diffusion and perfusion, and has been used to evaluate cell proliferation (23), oxygenation status (24), peritumoral infiltration (41), and histopathological features (42) in STS, with D predominantly contributing to these evaluations. Although previous studies

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have applied IVIM perfusion parameters in animal experiments to investigate sarcoma angiogenesis (25,26), few studies have examined their feasibility in clinical studies of STS. In the present study, D* and f showed moderate positive correlations with VEGF expression and MVD, consistent with previous studies of glioma (18), pancreatic cancer (22), and epithelial ovarian tumors (43). D* and f were significantly higher in tumors with high VEGF expression than in those with low VEGF expression. As VEGF expression and MVD increase in STS, the number of microvessels increases, leading to enhanced microcirculatory perfusion and an increased proportion of perfusion in the overall diffusion effect, resulting in increased D* and f values.

D reflects the restricted diffusion of water molecules and is inversely related to tumor cell density. In this study, no significant correlation was observed between D and VEGF expression or MVD, which is consistent with previous findings (25), indicating that D may not reflect the microvascular characteristics of STS. The b-value is crucial for IVIM-derived imaging parameters, and the number and range of b-values may contribute significantly to the inconsistent efficacy of clinical applications (44). A previous study (45) found that IVIM acquisition for soft-tissue tumors requires 6–8 b-values, including no fewer than four within the perfusion-sensitive range below 200 s/mm^2. In this study, five low b-values (0, 25, 50, 75, and 100 s/mm^2) were strategically selected under 200 s/mm^2. Since microcirculatory perfusion signal decays rapidly at low b-values, sufficient sampling within this interval can better characterize the fast diffusion component of intravoxel water movement, effectively constrain the nonlinear fitting of D* and f, and reduce fitting deviation caused by sparse sampling. Comparable low b-value protocols have previously yielded good reproducibility of D* and f (24,46), further supporting the rationale of our b-value scheme for reliable microcirculatory assessment in soft-tissue tumors.

Selecting appropriate perfusion imaging techniques is crucial in clinical practice. Although quantitative pharmacokinetic analysis of DCE-MRI data can accurately and intuitively describe microcirculatory characteristics, its application may be limited by complex pharmacological modeling and contraindications to contrast agents (36). The present study demonstrated that IVIM imaging and DCE-MRI have similar diagnostic capabilities for evaluating VEGF expression. Additionally, D* and Ktrans, as representative quantitative biomarkers of IVIM imaging and DCE-MRI, were identified as independent factors associated with VEGF expression and MVD. Further, D* and f were positively correlated with Ktrans and Kep, respectively, with D* and Ktrans showing the strongest correlation. These findings suggest that blood flow and vascular permeability may be the main factors affecting IVIM perfusion parameters.

IVIM is an effective, non-contrast medium-dependent technique that can noninvasively provide diffusion- and perfusion-related parameters and may serve as a supplementary sequence for reflecting detailed intratumoral pathological characteristics. However, given the rarity of STS, multiple histological subtypes were included in our cohort. A previous study (47) reported higher VEGF expression in undifferentiated pleomorphic sarcoma and leiomyosarcoma than in dermatofibrosarcoma protuberans, which is consistent with our findings. Such inter-subtype differences in angiogenic activity may confound correlation analyses between quantitative MRI parameters and angiogenic biomarkers. A single-subtype study design could reduce such confounding bias and better support targeted clinical assessment.

This study had a number of limitations. First, the relatively small sample size together with convenience sampling may have introduced selection bias. Given the heterogeneity of STS, the cohort may lack adequate representativeness, thereby limiting the generalizability of our findings. Second, the single-TE IVIM model adopted in this study does not fully account for differences in T2 relaxation times between the vascular and parenchymal compartments, which may lead to an overestimation of the f value (48,49). The multi-TE extended T2-IVIM model proposed by Jerome *et al.* (50) separates confounding T_2 effects from perfusion signals, eliminates the TE dependence of the f value, and yields more reliable f values. Nevertheless, their study also revealed that the coefficient of variation of single-TE IVIM (4.9–7.0%)

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was lower than that of multi-TE IVIM (23.4–35.0%), suggesting that the multi-TE model may trade repeatability for theoretical accuracy. Future studies should seek to implement multi-TE acquisition for analysis. Third, the D* and f values were relatively high in our study. This may be attributed to our ROI placement, which was selectively drawn on highly enhanced solid tumor regions to reflect angiogenic characteristics. Such targeted sampling inevitably introduces selection bias and elevates these perfusion-related parameters. Whole-tumor analysis rather than a hotspot ROI may reduce this sampling bias in future investigations. Fourth, the MRI-pathology control method is not suitable for abdominal, pelvic, or retroperitoneal masses, as access to deeper masses may be obstructed by multiple organs, making the method challenging to apply. Fifth, associations with clinical outcomes could not be assessed based on the relationship between MRI parameters and VEGF expression or MVD. Future multi-center studies should include patients with different STS subtypes and integrate the MRI-pathology control method with advanced artificial intelligence techniques, such as habitat analysis, to explore multi-dimensional pathological information and enhance its broader applicability.

Conclusions

Based on MRI-pathology image registration, this study demonstrated that DCE-MRI and IVIM imaging parameters can be used as markers for tumor angiogenesis in STS. Compared with other parameters, Ktrans and D* showed the greatest potential for evaluating VEGF expression and MVD in STS.

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Footnote

Reporting Checklist: The authors have completed the STARD reporting checklist. Available at https://qims.amegroups.com/article/view/10.21037/qims-2026-1-0086/rc

Data Sharing Statement: Available at https://qims.amegroups.com/article/view/10.21037/qims-2026-1-0086/dss

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The clinical study was approved by the Ethics Committee of The Second Hospital of Dalian Medical University (No. KY2025-069-01). Written informed consent was signed by all participants in this study.

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